MILITARY SPECIFICATION

PRIMER, ELECTRIC, M80A1 (ALTERNATE) LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the loading, assembling and packing of Primer, Electric, M80A1 (Alternate).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-P-223	- Powder, Black
MIL-E-46215	- Element, Ignition, Electric Metal Parts Assembly, For Igniter, Electronic Primer, M63 and Electric Primers, M80A1 and M83
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286 - Propellants: Sampling, Inspection and Testing
MIL-STD-1234 - Pyrotechnics, Sampling, Inspection and Testing

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

8837899 - Carton Packing, Ammunition, For Primer, Electric, M80A1

8837900 - Box, Packing, Ammunition, for Primer, Electric, M80A1

8839499 - Primer Electric, M80A1 Sheet 2 - Loading Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Primers.-The loaded primers shall comply with all requirements specified on Drawing 8839499 (Sheet 2) and with all requirements specified in applicable specifications.

3.2 Parts.-The parts shall comply with all requirements specified on drawings listed in paragraph 2.1 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Moisture content

3.3.1 Ignition element charge composition.-The moisture content of the ignition element charge composition at the loading station at the time of loading the ignition element shall not exceed 0.10 percent.

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3.3.2 Black powder.-The moisture content of the black powder at the time of loading the closing plug shall not exceed 0.50 percent.

3.3.3 Benite.-The moisture content of the benite at the loading station at the time of loading shall not exceed 1.00 percent.

3.4 Electrical resistance test.-The head loading assembly and the loaded primers, shall have an electrical resistance of not less than 0.80 ohms nor more than 1.80 ohms.

3.5 Functioning/non-functioning test (Head loading assembly)

3.5.1 Functioning.-The head loading assembly shall function when a current of 1.25 amperes, maximum, is applied.

3.5.2 Non-functioning.-The head loading assembly shall not function when a current of 190 to 200 milliamperes is applied.

3.6 Lacquer in flash holes.-The lacquer in primer flash holes shall comply with illustration 1, 2 or 3 of Figure 1.

3.7 Proving ground.-The primer shall comply with the following requirements when fired ballistically:

a. The primer shall not fail to fire.
b. The primer body shall not rupture or become detached.
c. There shall be no audible hangfire(s), (See 6.6.a).
d. The primer shall not squib (See 6.6.b).

3.8 Torque requirements

3.8.1 Assembly torque.-Threaded joints shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.8.2 Disassembly torque.-Threaded joints shall withstand application of the minimum disassembly torque specified on the applicable drawing, without moving, after thread sealing compounds has cured.

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3.9 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship.-All parts and assemblies shall be fabricated and loaded in accordance to the applicable specifications and drawings. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part of assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted at this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated in Table I.

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Part/Assembly Description	Drawings	Quantity
Element, Ignition, Electric, Loading Assembly	8839505	10
Plug, Closing Loading Assembly	9327346	10
Head Loading Assembly	9327345	400
Primer Electric, M80A1 Loading Assembly	8839499 (Sheet 2)	100
Carton, Packing, Ammunition for Primer, Electric, M80A1	8837899	5
Box, Packing, Ammunition for Primer Electric, M80A1	8837900	5

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Primer Assembly & Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Element, ignition, electric loading assembly (Dwg. 8839505)				
	Examination for defects	10			4.4.2.1
	Plug, closing loading assembly (Dwg. 9327346)				
	Examination for defects	10			4.4.2.2
	Head loading assembly (Dwg. 9327345)				
	Examination for defects	400			4.4.2.3 & 4.4.2.4
	Functioning	200*			4.5.4
	Non-functioning	200*			4.5.5
	Primer, electric, M80A1 loading assembly (Dwg 8839499) - sheet 2				
	Examination for defects	50**			4.4.2.5
	Proving ground	50**			4.5.7
	Carton, packing, ammunition for primer, electric, M80A1				
	Examination for defects	5			4.4.2.6 & 4.4.2.7
	Box, packing, ammunition for primer, electric, M80A1				4.4.2.8/4.4.2.9
	Examination for defects	5			4.4.2.10

NOTES: *Should any sample fail to meet the applicable requirements, the First Article Sample shall be rejected.
**If a critical defect occurs or if 2 or more major defects occur, the First Article Sample shall be rejected.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded primers shall contain:

a. Metal parts assemblies for electric ignition element from one lot interfix from one manufacturer.
b. Ignition element charge composition produced by one supplier under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit of chemical or physical mixing process intended to made the final product homogeneous.
c. Black powder from not more than one lot from one supplier.
d. Primer metal parts from one lot interfix from one manufacturer.
e. Benite of one lot interfix from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Element, ignition, electric, loading assembly				8839505
					NEXT HIGHER ASSEMBLY
					9327345
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Charge above flush		100%	3.2	Straight-edge
2	Charge composition on outside surface		100%	3.2	Visual
Major					
101	Charge missing or insufficient		0.40%	3.2	Visual
102	Moisture content	See Note 1		3.3.1	4.5.2.1
103	Analysis of ignition element charge composition	See Note 2		3.2	4.5.6
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

Note 1: One (1) sample, minimum, shall be selected from each eight (8) hours production. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected.

Note 2: Five (5) grams of ignition element charge composition shall be selected from that quantity of composition used in loading a complete lot of head loading assemblies.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Plug, closing loading assembly				9327346
					NEXT HIGHER ASSEMBLY 9327345
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Black powder charge missing		100%	3.2	Visual
102	Disc(s) missing, punctured or peeling (when applicable)		0.40%	3.2	Visual
103	Lacquer missing from disc (when applicable)		0.40%	3.2	Visual
104	Tape missing or improperly assembled (when applicable)		0.40%	3.2	Visual
105	Moisture content of black powder charge	See Note		3.3.2	4.5.2.2

Note: One (1) sample, minimum, shall be selected from each eight (8) hours production. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Head loading assembly, prior to assembling closing plug assembly		9327345
			NEXT HIGHER ASSEMBLY
			8839499 (sheet 2)

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Closing disc missing or damaged to the extent that explosive is exposed		0.40%	3.2	Visual
102	Retainer fails to withstand application of minimum assembly torque		0.40%	3.8.1	4.5.8.1
103	Retainer stake missing (when applicable)		0.40%	3.2	Visual
104	Lacquer missing from retainer threads (when applicable)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Head Loading Assembly				9327345
					NEXT HIGHER ASSEMBLY
					8839499 (Sheet 2)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non-Functioning test	See 4.4.3.1.2		3.5.2	4.5.5
Major					
101	Functioning test	See 4.4.3.1.1		3.5.1	4.5.4
102	Electrical resistance test		100%	3.4	4.5.3
103	Depth to electrode		0.40%	3.2	Gage
104	Flash hole obstructed		0.40%	3.2	Gage
105	Retainer fails to withstand application of minimum assembly torque. (See note if lacquer is used)		0.40%	3.8.1	4.5.8.1
106	Closing plug fails to withstand application of minimum assembly torque (See note)		0.40%	3.8.1	4.5.8.1
Minor					
201	Lacquer missing from closing plug threads		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTE: Torque test shall be performed while lacquer is still wet (within 5 minutes after the lacquer applicaton).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63313 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Primer, Electric, M80A1				8839499 (sheet 2)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proving ground	See 4.4.3.2		3.7	4.5.7
Major					
101	Liner improperly positioned (prior to loading)		0.40%	3.2	Visual
102	Flash hole sealing improper (prior to loading)		0.40%	3.6	Visual
103	Weight of benite charge		0.40%	3.2	Scale
104	Length of benite charge		0.40%	3.2	Gage
105	Lacquer missing from plug threads		0.40%	3.2	Visual
106	Sealant missing from body threads		0.40%	3.2	Visual
107	Body fails to withstand minimum disassembly torque (after curing)		0.40%	3.8.2	4.5.8.2
108	Moisture content of benite	See note		3.3.3	4.5.2.3
109	Proving ground	See 4.4.3.2		3.7	4.5.7
110	Electrical resistance test		100%	3.4	4.5.3
111	Length, total, max.		0.40%	3.2	Gage
Minor					
201	Depth to plug		0.65%	3.2	Gage
202	Marking incorrect or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:
One 100 gram sample of benite shall be selected at the loading station, during each 8 hours production. If the moisture content of the sample exceeds the applicable requirement the primers loaded since the last acceptable sample shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Carton, prior to sealing				8837899
					NEXT HIGHER ASSEMBLY
					8837900
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any packing material missing		0.40%	3.2	Visual
102	Separator(s) damaged to the extent that primers touch each other		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837899
4.4.2.7	Carton, sealed				NEXT HIGHER ASSEMBLY 8837900
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton completely		0.65%	3.2	Visual
202	Carton torn, cut or punctured		0.65%	3.2	Visual
203	Contents loose to an extent greater than permitted by packing design		0.65%	3.2	Manual
204	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Sealed barrier bag				8837899
					NEXT HIGHER ASSEMBLY
					8837900
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag punctured, torn or cut		0.40%	3.2	Visual
102	Seal mislocated or incomplete		0.40%	3.2	Visual
103	Adequacy of heat seal	See Note		3.2	Test
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTE: Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Box, packing, prior to sealing				8837900
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Carton missing		0.40%	3.2	Visual
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63313(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Box, packing sealed				8837900
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protruding nail		0.40%	3.2	Visual
102	Board broken or loose		0.40%	3.2	Visual/Manual
103	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Strapping improperly engaged or mislocated		0.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Head loading assembly

4.4.3.1.1 Functioning

4.4.3.1.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the requirements specified, five hundred (500) samples shall be selected from each lot for test as specified in 4.5.4. If four (4) or more failures occur, the lot shall be rejected. The equipment used shall be in accordance with 4.4.4. (Destructive test)

4.4.3.1.1.2 After three consecutive lots have complied with the criteria of 4.4.3.1.1.1, a double sampling plan in accordance with Code Letter L and an AQL of 0.25 percent shall be used for each lot. The test shall be performed as specified in 4.5.4 using equipment specified in 4.4.4. (Destructive test)

4.4.3.1.2 Non-functioning

4.4.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the specified requirement, two hundred (200) samples shall be selected from each lot and tested in accordance with 4.5.5 using equipment specified in 4.4.4. If any assembly functions, the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot. (Destructive test)

4.4.3.1.2.2 After three consecutive lots have complied with the criteria of 4.4.3.1.2.1 eighty (80) samples shall be selected from each lot for this test. If any assembly functions the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot. (Destructive test)

4.4.3.2 Proving ground - (See 3.7).-The primer shall be observed for evidence of the following defects:

TABLE II

Defects	Classification
Body ruptures or becomes detached	Critical
Failure to fire	Major
Assembly hangfires	Major
Assembly squibs	Major

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4.4.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, thirty-five (35) primers shall be randomly selected from each lot for this test. If a critical defect occurs or one (1) or more major defects occur, the lot shall be rejected.

4.4.3.2.2 After three (3) consecutive lots have met the criteria specified in 4.4.3.2.1, ten (10) primers shall be randomly selected from each lot for test. If one (1) critical defect occurs or if one (1) or more major defect occur, the lot shall be rejected.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Moisture content

4.5.2.1 Igniter charge composition.-The moisture content of the igniter charge composition shall be determined as specified in Method 101.5 of MIL-STD-1234 except that a five (5) gram sample shall be used.

4.5.2.1.1 Alternate method.-The moisture content of the igniter charge composition may be determined in accordance with the electrolytic hydrometer described in Method 101.4 of MIL-STD-286 using a sample size not greater than 0.2 grams and a temperature of 70 degrees Centigrade.

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4.5.2.2 Black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in MIL-P-223.

4.5.2.3 Benite.-The moisture content of the benite shall be tested in accordance with Method 102.1 of Standard MIL-STD-286.

4.5.3 Electrical resistance.-The head loading assembly (dwg. 9327345) and the primer assembly (dwg. 8839499 - sheet 2), shall be tested for electrical resistance by applying an electrical current or twenty (20) milliamperes, maximum, using the electrode as one contact point and the head as the other. The equipment shall be in accordance with 4.4.4.

4.5.4 Functioning.-The head loading assembly shall be tested for electrical functioning by applying an electrical current of 1.25 amperes, maximum, using the electrode for one contact point and the head for the other. The equipment shall be in accordance with 4.4.4.

4.5.5 Non-functioning.-The head loading assembly shall be tested for non-functioning by applying an electrical current of one hundred and ninety (190) to two hundred (200) milliamperes for one second, minimum, using the electrode as one contact point and the head as the other. The equipment shall be in accordance with 4.4.4.

4.5.6 Determination of ignition charge composition.-The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate	44.5 plus or minus 2
Lead Thiocyanate	35.5 plus or minus 2
Charcoal	20.0 plus or minus 1.5
Lacquer Solids (added)	1.6 plus or minus 0.2

4.5.6.1 Lacquer solids content.-Transfer approximately 1.5 grams of the sample to a tared medium porosity filtering crucible and allow to air dry for one hour. Place the crucible and contents in a vacuum oven maintained at 55 plus or minus 2 degrees C. for 2 hours, cool in a desiccator and weigh. Repeat this procedure until a constant weight within 0.5 milligrams is attained. The content of the crucible shall then be extracted with ten 25 ml. portions of acetone saturated with potassium chlorate and lead thiocyanate. The crucible and content shall be aspirated until the odor of acetone is no longer perceptible and shall then be dried in a vacuum oven maintained at 55 plus or minus 2 degrees C. for 2 hours, cooled in a desiccator and weighed. The crucible and contents shall be used for the charcoal determination. The percent lacquer solids shall be calculated as follows:

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$$\text{Percent lacquer solids} = \frac{(A-B) \times 100}{B}$$

Where:
A = weight of the dried sample
B = weight of the dried sample after acetone extraction

4.5.6.2 Charcoal content.-With the aid of a Fisher High Form Filtrator, or equivalent, the contents of the crucible remaining after the lacquer solids determination shall be extracted with ten 25 ml. portions of 5 percent potassium nitrate solution followed by 50 ml. of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Aspirate the residue for several minutes, place in an oven maintained at 100 plus or minus 5 degrees C. for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{A \times 100}{B}$$

Where:
A = weight of sample after acetone and potassium nitrate extraction
B = weight of sample after acetone extraction

4.5.6.3 Lead thiocyanate.-Exactly 35 ml. of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml. of 3N nitric acid (nitric acid used shall be water-white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for 5 minutes. Five (5) ml. of a saturated solution of ferric ammonium sulfate shall be added and the solution titrated with standardized 0.1 N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

Where:
A = volume of silver nitrate solution used
B = normality of silver nitrate solution
C = volume of potassium thiocyanate solution
D = normality of potassium thiocyanate solution
E = weight of sample after acetone extraction (see 4.5.6.2)

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4.5.6.4 Potassium chlorate content.-The percent of potassium chlorate is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent.

4.5.7 Proving ground test.-The primers shall be assembled into complete rounds for which the primer is standard and fired from an M60 or M60A1 tank (or from any appropriate facility) with an M68 gun in accordance with ATP IS-P-17J. The test will be valid if all rounds fired have piezo equivalent pressures between 65,000 psi (448.16 MPa) and 71,100 psi (490.22 MPa). If any round fired has a pressure below the minimum, the lot shall be referred. If any round fired has a pressure above the maximum, the lot shall be accepted provided no ballistic failures have occurred. Should failures occur on a round which exceeds the maximum pressure, the lot shall be referred (see 6.7).

4.5.8 Torque testing

4.5.8.1 Head loading assembly.-Whenever lacquer is used, the torque test shall be performed while the lacquer is still wet (within 5 minutes after the lacquer application). The torque shall be applied in a tightening direction, while holding the head. The joint under test shall be marked whenever is possible to provide a means of detecting any movement between adjacent parts. Any discernable movement between the adjacent parts shall be regarded as a torque test failure.

4.5.8.2 Primer loading assembly.-After a 2 hour minimum curing of sealant, the torque shall be applied in the loosening direction, while holding the head. The joint under test shall be marked to provide a means of detecting any movement between adjacent parts. Any discernable movement between adjacent parts shall be regarded as a torque test failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging and marking

5.1.1 Level B.-The primers shall be packaged and marked in accordance with dwg. 8837899.

5.2 Packing and marking

5.2.1 Level B.-The packaged primer shall be packed and marked in accordance with dwg. 8837900.

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6. NOTES

6.1 Intended use.-The primer covered by this specification is intended for use on 105MM Cartridges, APDS-T M392, APDS-T M728, TPDS-T M724.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals - See MIL-A-48078.-Submit designs as required to Commander, US Army Armament Reserach and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Reserach and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Check test for possible deterioration of primers.-The following tests are designed to prevent loading major ammunition items with primers which have deteriorated. The tests should be performed on each lot of M80A1 Primers which have been stored for more than two years or which have been subjected to any adverse conditions, however brief, within that period of time. If the specification for the ammunition item into which the M80A1 Primer is assembled, does not require the performance of the following check test, the contracting officer shall contractually provide for compliance of check test requirements. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

6.5.1 Moisture content of benite.-A sample of 25 primers shall be selected for this test. If one or more primers fail to comply with the requirements specified in 3.3.3, the lot shall be rejected. The test shall be performed in accordance with Standard MIL-STD-286, Method 102.1.

6.5.2 Electrical resistance.-This test shall be performed 100 percent. Any primer assembly which fails to comply with the requirement specified in 3.4 shall be classes defective and removed from the lot. The test shall be performed as specified in 4.5.3.

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PRIMERS - STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

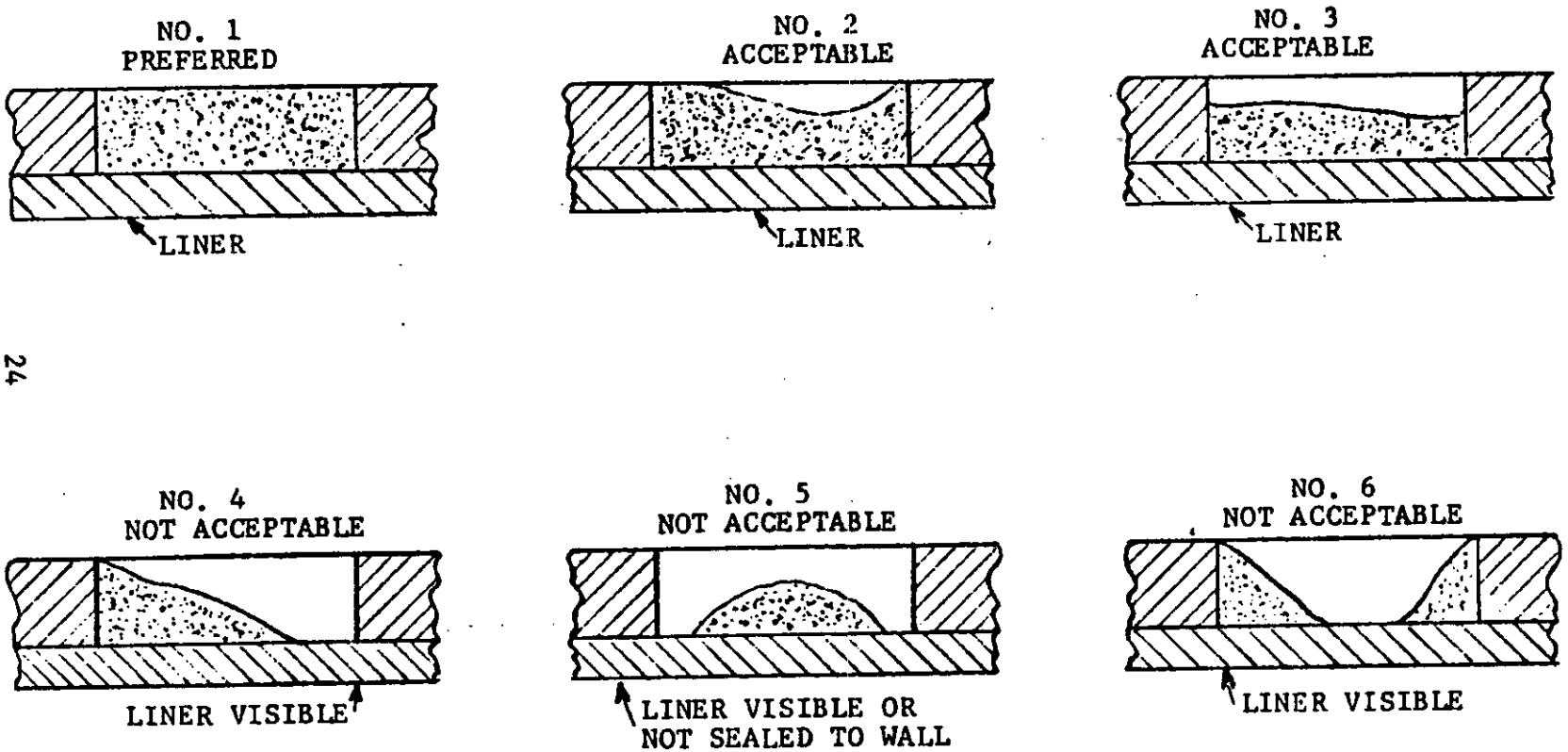


FIGURE 1

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6.5.3 Functioning (head loading assembly).-300 primers shall be selected from each lot for this test. The head loading assemblies shall be removed from the primers and tested as specified in 4.5.4. If three of more assemblies fail to comply with the applicable requirement the lot shall be rejected.

6.5.4 Non-functioning (head loading assemblies).-50 primers shall be selected from each lot for this test. The head loading assemblies shall be removed from the primers and tested as specified in 4.5.5. If one or more assemblies fail to comply with the applicable requirement the lot shall be rejected.

6.6 Definitions:

a. Hangfire.-A perceptible delay occuring between the instant application of energy and the functioning of primer/propellant.

b. Squib.-A low order explosive or slow rate burning as exhibited by no functioning of the propellant charge.

6.7 Assessment of primer for metal parts security.-The intent of this specification is to test the primer at the highest pressure expected in the field. This primer is used on the M392, M728 and M724 cartridges. Any of these cartridges may be used as the test vehicle during ballistic acceptance testing of the primer.

6.8 Submission of test data.-In addition to the normal distribution of records, when the primers are procured by the US Army Materiel Readiness Command, one copy of all ballistic test data shall be forwarded to each of the following:

Commander
US Army Research and Development Command
ATTN: DRDAR-QAR-E and DRDAR-LCU-CV
Dover, NJ 07801

Commander
ARRCOM
ATTN: DRSAR-QAD
Rock Island, IL 01201

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project No.: 1390-A203

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63313 Primer Electric, M80A1 (Alternate) LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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DATE

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